

The variance report is green and the money is gone

An analyst case study — Lumen Optics Manufacturing photonics spend

Reporting date: 2025-12-31 · **Tooling:** T-SQL (SQL Server), Power BI (DAX), Excel

Data disclosure. Lumen Optics Manufacturing is a fictional photonics manufacturer. Every figure in this study is synthetic, generated by the scripts in `sql/`, and reproducible from them. No confidential data is used and no claim is made about any production system.

1. The situation

Lumen Optics Manufacturing builds fibre-coupled laser modules. It buys around **\$48.6M** a year of direct materials — laser diodes, photodetectors, optical fibre, lens assemblies, coatings, precision mounts, connectors and thermal control — from 36 suppliers across 140 part numbers.

Procurement reports purchase price variance monthly. It has been green all year: **-0.19%**, meaning Lumen pays slightly *less* than its contracts require. Every category manager is hitting their number.

Material cost per module has risen anyway. Nobody can say why.

The question

Our purchase prices are flat or falling and every variance report is green — so why is material cost per accepted unit rising, and which supplier-part combinations should sourcing renegotiate first?

2. Why the variance report cannot answer it

Purchase price variance compares what was paid against what was **agreed**. That makes it structurally blind to the largest source of loss in a photonics book.

✦ **Optical components follow learning curves.** As cumulative volume rises and processes mature, unit costs fall — laser diodes at roughly 8% a year, connectors 7%, coatings 5%, fibre 4%. This is not a forecast; it is how the category behaves.

A supplier who renews a contract **flat** while its category erodes 8% a year keeps everything the curve was supposed to hand back. And because the price never *rose*, purchase price variance records zero. Forever.

The measure that sees it is **erosion capture**: the share of the expected category decline that actually showed up in the prices paid.

Expected erosion, spend-weighted across the book	benchmark
Erosion actually captured	29.50%
Target	80.00%
Annual value of the gap	\$4,047,609

Lumen is collecting under a third of what its categories are giving away. That is roughly **8.3% of annual direct-materials spend**, and no report in the business currently shows it.

3. Approach

3.1 One definition of what a purchase cost

$$\text{LandedCost} = \text{OrderQty} \times \text{UnitPrice} + \text{FreightAmount} + \text{ExpediteFee}$$
$$\text{CostPerAcceptedUnit} = \text{LandedCost} \div \text{QtyAccepted}$$

Freight and expedite are held separately from the unit price on purpose: a unit price is negotiated with a supplier, an expedite fee is caused by planning, and folding them together hides which of the two is costing the money.

The second line is the one that changes decisions — see §4.2.

3.2 A price agreement is a dated fact, not an attribute

A contracted price is true between two dates and false outside them. Storing it on the part or the vendor would let today's contract retroactively govern a purchase made two years ago, so every historical variance would be measured against a price that did not exist at the time.

Two consequences the model has to handle explicitly:

- **An open-ended agreement is still in force.** Excluding `NULL` end dates is the usual way contract coverage is made to look far worse than it is. 1,333 lines here are governed by one of 187 open agreements (UAT-03).
- **Overlapping agreements make the contracted price ambiguous.** 61 lines have two agreements in force at once. The resolution is deterministic by construction — latest start, then highest key — and the ambiguity is *reported* rather than silently resolved, because a variance computed off a coin-flip is a number with no error bar.

3.3 Every number computed twice

Each headline figure is implemented independently in T-SQL and again in Excel formulas, then reconciled on the workbook's Validation sheet in the open. **All 14 checks agree**, and the workbook says so on its own face.

4. What the data says

4.1 The erosion gap is concentrated, and it is where leverage is weakest

Of 185 vendor–part pairs with enough history to judge:

Erosion band	Pairs	Spend (12m)	Share	Opportunity
Beats the benchmark	19	\$4,255,931	9.7%	—
Tracks it (60–100%)	45	\$9,396,584	21.4%	\$317,630
Partial (20–60%)	60	\$12,712,195	29.0%	\$1,092,403
Flat (under 20%)	61	\$17,495,221	39.9%	\$2,637,575

Two fifths of the spend sits with suppliers capturing less than a fifth of the expected erosion. But the sharper cut is by leverage:

	Pairs	Opportunity	Avg leverage score	Erosion captured (spend-weighted)
Parts with 2+ qualified suppliers	152	\$2,576,657	72.8	38.26%
Parts with one qualified supplier	33	\$1,470,952	38.1	8.91%

Sole-source suppliers capture 8.91% of the expected erosion; competitive ones capture 38.26%.

Suppliers hold price precisely where Lumen cannot leave.

Both figures are **spend-weighted**, on the same basis as the 29.50% headline. An earlier draft of this table quoted simple averages (37.77% and 10.88%), which are not comparable to the headline and understated the gap — weighting by money widens it from 3.5× to 4.3×. Across 185 pairs ranging from under \$4,000 to over \$1 million in trailing-twelve-month spend, an unweighted average lets a small pair count as heavily as one carrying nearly three hundred times the money.

That is not a surprise economically — it is what anyone would predict. What matters is that it is now **quantified**: 36% of the total opportunity sits behind parts where the negotiating position is half as strong, and treating those two populations identically in a sourcing plan wastes the year.

4.2 The cheapest price was the dearest usable part — until the price fell further

VEN-004 undercuts every competitor on laser diodes, on all three parts it shares with them, in every year. It also rejects far more of what it ships: on LOM-0072, 8.4% against 1.0% for the other two suppliers — 8.4 times the rate. Whether it is the cheapest *usable* supplier therefore depends on which is moving faster, its price or its rejects, and on these parts the answer changed.

Part LOM-0072, three qualified suppliers, **cost per accepted unit** by order year:

Vendor	2023	2024	2025	All history	Avg unit price	Acceptance
VEN-004	\$227.76 (dearest)	\$192.58 (cheapest)	\$168.28 (cheapest)	\$204.71	\$184.14 (cheapest)	91.61%
VEN-012	\$212.00	\$204.35	\$189.42	\$204.44	\$197.43	98.99%
VEN-009	\$223.59	—	\$193.55	\$213.94	\$211.49	99.09%

In 2023 VEN-004 was the cheapest per unit and the **dearest per usable component**: a 2% price advantage did not cover a 9% rejection rate, and on every shared part it cost more per accepted unit than the best alternative. Its prices then fell between 11% and 16% a year, far faster than its rejection rate moved, and in 2024 and 2025 it was the **cheapest per accepted unit on all three shared parts** — on LOM-0072 in 2025, by \$21 a unit. Over all history the two effects nearly cancel: \$204.71 against VEN-012's \$204.44 is a blend of one bad year and two good ones, not a finding.

Two lessons, and neither is "drop the cheap supplier". Compare suppliers on **cost per accepted unit**, never unit price: unit price got 2023 wrong. And compare them over a **recent window**, not all history: all history calls 2025 a 27-cent tie. The quality problem is still real — VEN-004 accounts for **\$289,024 (18%)** of the material Lumen paid for and could not use — and fixing it would make the best-value supplier on these parts cheaper still.

A component that arrives outside specification was still paid for, still freighted, and cannot be built into a module. Unit price cannot see any of that; cost per accepted unit is the only comparison that can.

Across the book this costs **\$1,571,168** of material paid for and unusable.

4.3 A quarter of the spend has nothing to negotiate against

Maverick spend: 27.09% — against a 5% target and a 12% warning.

The cause is not rogue buying; it is contracts quietly expiring. Roughly one agreement in seven is never renewed, and purchasing simply continues at drifting spot prices. Maverick spend and the erosion failure turn out to be the same story: a lapsed contract is a price nobody is watching.

There is nothing to renegotiate until there is something to renegotiate, which is why `PUT_ON_CONTRACT` is a distinct action in the queue carrying **\$1,623,978** of opportunity across 48 pairs.

4.4 Portfolio figures hide the cohorts that are failing

Metric	Portfolio	Status	The cohort underneath
Acceptance rate	99.11%	Green	VEN-004 on Laser Diode: 91.4%
On-time delivery	80.62%	Red	VEN-009: 20.8% of 2025 orders on time, against 83% in 2023
Expedite spend	0.23%	Green	Doubled between 2023 and 2025 across the book (0.12% → 0.23%)

The acceptance figure is the instructive one. At vendor level VEN-004 sits at 98.3% and looks unremarkable; only the **vendor × category** cut exposes 91.4%. A portfolio average cannot breach until the whole book does, and by then the cohort that caused it has been failing for a year.

Every aggregate metric in this project therefore ships with its cohort cut attached to the same report.

5. The operational control

185 vendor–part pairs is not a plan. A category manager can run perhaps eight serious negotiations a quarter, so the queue stays complete for audit and marks what is actually workable.

43 pairs this quarter, carrying 66.1% of the \$4.05M opportunity.

Action	Pairs	Opportunity	Spend (12m)	This quarter
RENEGOTIATE	54	\$1,631,840	\$16,221,031	15
PUT_ON_CONTRACT	48	\$1,623,978	\$11,559,724	20
DUAL_SOURCE	8	\$515,330	\$4,652,024	8
ACCEPT	73	\$276,461	\$10,320,429	0
FIX_QUALITY	2	—	\$1,106,723	0

Four design decisions are worth naming:

- **Ranking is by money; the ACTION is by leverage.** Per pair, the largest gaps sit on sole-source parts where there is no credible threat to make. Ranking on value alone sends a category manager into meetings they cannot win, which is the fastest way to lose a sourcing team's trust in a report.
- **Quality outranks price.** If the loss is rejected material rather than price, a price conversation is the wrong conversation. `FIX_QUALITY` routes those to a corrective action.
- **Above \$100,000 a year, ask anyway.** Weak leverage is a reason to expect a hard conversation, not a reason to skip one. Before this rule existed, a \$179k annual gap was classified `ACCEPT` because the supplier was large enough not to care.

- **ACCEPT is an honest answer.** 73 pairs worth \$276k in total — averaging under \$4k each — are parked with a reason. Saying so is more useful than listing them forever.

6. Recommendations

#	Recommendation	Owner	Basis	Value
1	Work the quarter's 43 pairs, starting with the 20 PUT_ON_CONTRACT	Category managers	43 pairs carry 66.1% of the total gap	\$2,673,946 of the \$4.05M
2	Re-paper the lapsed agreements before negotiating anything on them	Sourcing ops	27.09% of spend has no agreement in force	Unlocks \$1,623,978 that cannot be negotiated today
3	Add erosion capture to the monthly pack, beside purchase price variance	CPO	PPV reads -0.19% Green while \$4.05M walks out	Makes the invisible loss visible monthly
4	Keep VEN-004 on laser diodes, and put its rejection rate under corrective action	Optoelectronics, with supplier quality	Cheapest per <i>accepted</i> unit on all three shared parts in 2024 and 2025, but rejecting 8.4 times as often as its competitors	\$289,024 of the \$1,571,168 of unusable material is VEN-004's
5	Qualify a second source on the 8 DUAL_SOURCE pairs	Category managers	Sole-source capture is 8.91% against 38.26% where alternatives exist; these 8 carry at least \$40,000 a year and can be requalified within 8 months	\$515,330 , all 8 scheduled this quarter
6	Put VEN-009 on a delivery corrective action	Supplier quality	On time for 83% of 2023 orders and 20.8% of 2025's (59.64% over all history)	The worst delivery record of the 36 vendors
7	Report every aggregate with its cohort cut	Analytics	Acceptance is 99.11% company-wide and 91.4% for one vendor-category	Stops green lights hiding failing cohorts
8	Resolve the 6 overlapping agreements	Sourcing ops	Contracted price — and therefore PPV — is ambiguous on 61 PO lines, \$1,495,917 of spend	Makes the variance figure meaningful

Recommendations 2 and 3 cost nothing but a decision. Recommendation 1 is what the sourcing team does on Monday.

7. What went wrong during the build

Five defects reached a working state before being caught. How each was found matters more than that it existed.

7.1 A planted pattern that silently did not exist

The generator was told to make one named vendor the cheap-but-rejected laser diode supplier. But vendor-part assignment is random, and that vendor was never assigned a single laser diode. The `UPDATE` matched zero rows — **which does not raise an error**. It simply did nothing, and the pattern was absent from the first profiling run.

Now the target is *derived* from the data (`the vendor supplying the most laser diode parts`) and stored where both the price logic and the rejection logic read it.

7.2 A lapse that produced an eternal contract

Non-renewal is simulated by deleting the later contract terms. The surviving final term was then marked open-ended — so a contract that **lapsed** became a contract that **never expires**, the exact opposite of the pattern.

The tell was not an error: off-contract spend stayed at exactly 222 lines across two runs while the lapse rate was more than doubled. A number that refuses to move when its driver changes is evidence of a broken dependency.

7.3 A window function that saw one row

A pair-level erosion query reported *"185 of 185 vendor-part pairs holding price flat — 100%"*, directly contradicting the category table above it, which showed prices falling.

The cause was `ROW_NUMBER() OVER (PARTITION BY ...)` inside a `CROSS APPLY` : the window function evaluates over the **single row** it was applied to, so every row received rank 1 both ascending and descending, making first price and last price the same number.

A tidy way to produce a confident, plausible, entirely wrong headline.

7.4 Published figures that disagreed with their own components

`ExpectedPrice` was computed from an unrounded exponent while `YearsElapsed` was published rounded to three decimals. A reader redoing the arithmetic from the printed numbers got a different answer than the report. The same applied to erosion capture.

The rule now applied in SQL **and** in the DAX and Excel implementations: **a derived figure is computed from its components as published, not from their unrounded intermediates**. UAT-05 and UAT-07 assert it.

7.5 One defect reported twice, and one test that was simply wrong

Duplicated PO lines are inserted after receipts are generated, so they can never have a receipt — and every one was also being reported as `NEVER_RECEIVED`, padding the defect count with a consequence of something already on the report. Excluding them dropped the count from 114 to 94.

Separately, three UAT cases failed on their own premises rather than on the code: one compared a rounded published figure against an unrounded recomputation, one compared committed spend against received-only spend, and one held a constant that a later fix had moved. **The test failing does not mean the code is wrong** — and the instinct to "fix" the code first is how a correct implementation gets broken to satisfy a bad assertion.

8. What I would do differently

- **Derive planted patterns from the data, never name them.** §7.1 cost a full profiling cycle, and the failure mode — a zero-row `UPDATE` — is silent by design in SQL.
- **Treat "a number that did not move" as a finding.** §7.2 was caught only because I compared two runs. A generator change that does not change its output is either a no-op or a broken dependency, and both deserve investigation before the data is used.
- **Build the should-cost model alongside the erosion model.** Erosion measures the failure to decline *from the first observed price*. It cannot say whether that opening price was fair — a part bought badly in 2023 and eroded perfectly since looks healthy here. That is a real limitation and the case for the complementary analysis, not a caveat to bury.
- **Put the leverage cut in front of the opportunity ranking, not beside it.** The most useful single table in this study is §4.1's sole-source comparison, and it emerged from a review query rather than from the designed output. It should have been a headline from the start.

9. How to verify any figure in this study

```
sqlcmd -S <server> -E -d master      -i sql/01_create_schema.sql
sqlcmd -S <server> -E -d master      -i sql/02_generate_synthetic_data.sql
sqlcmd -S <server> -E -d LumenSpend -i sql/03_core_views.sql
sqlcmd -S <server> -E -d LumenSpend -i sql/04_data_quality_checks.sql
sqlcmd -S <server> -E -d LumenSpend -i sql/05_kpi_views.sql
sqlcmd -S <server> -E -d LumenSpend -i sql/06_stored_procedures.sql
sqlcmd -S <server> -E -d LumenSpend -i sql/07_uat_test_cases.sql      # 21 cases

EXEC dbo.usp_SpendScorecard      @AsOf = '2025-12-31';
EXEC dbo.usp_RenegotiationQueue  @AsOf = '2025-12-31', @ThisQuarterOnly = 1;
EXEC dbo.usp_PriceErosionDetail  @AsOf = '2025-12-31', @MinOpportunity = 50000;
EXEC dbo.usp_VendorScorecard     @AsOf = '2025-12-31';
```

```
EXEC dbo.usp_CategorySummary      @AsOf = '2025-12-31', @GroupBy = 'Category';
EXEC dbo.usp_PartPriceHistory     @PartNumber = 'LOM-0072';
EXEC dbo.usp_RunDataQualityChecks;

powershell erp_extracts/Export_Extracts.ps1
powershell excel/Build_Workbook.ps1
powershell excel/Validate_Workbook.ps1                                     # 14 checks
```

The generator is deterministic — hash-keyed rather than `RAND()` — so a rebuild reproduces the dataset byte for byte, and UAT-20 asserts the fingerprints.

Appendix — headline figures at 2025-12-31

Metric	Value	Target	Warning	Status
Erosion capture	29.50%	80.00	60.00	Red
Maverick spend	27.09%	5.00	12.00	Red
Purchase price variance	−0.19%	0.00	2.00	Green
Acceptance rate	99.11%	98.00	95.00	Green*
On-time delivery	80.62%	95.00	90.00	Red
Expedite spend	0.23%	1.00	3.00	Green
Single-source spend	27.17%	20.00	35.00	Amber
Top-5 vendor share	27.20%	50.00	70.00	Green

* Green at portfolio level only. VEN-004 on Laser Diode reads 91.4%.

Direct-materials spend (12 months)	\$48,586,386.86
Landed cost (12 months)	\$49,474,785.92
Annual erosion opportunity	\$4,047,609
of which on sole-source parts	\$1,470,952
Value already ahead of the curve (not netted)	−\$249,429
Material paid for and unusable	\$1,571,168
Vendor–part pairs assessed	185
Workable this quarter	43 (66.1% of the gap)